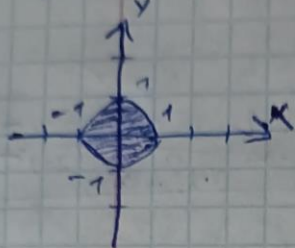


$$P_{360}/E \times 2.$$

$$y = 1 - x^2$$

$$y = 0$$



$$R = 1 - x^2$$

$$A = \pi (1 - x^2)^2$$

$$\int_{-1}^1 \pi (1 - x^2)^2 dx =$$

$$= \pi \int_{-1}^1 (1 - 2x^2 + x^4) dx =$$

~~$$= \pi (x - 2x^3 + x^5) \Big|_{-1}^1$$~~

$$= \pi \left(x - \frac{2}{3}x^3 + \frac{x^5}{5} \right) \Big|_{-1}^1 =$$

$$= \left(\pi - \frac{2}{3}\pi + \frac{1}{5}\pi \right) - \left(-\pi + \frac{2}{3}\pi - \frac{1}{5}\pi \right) =$$

$$= 2\pi - \frac{4}{3}\pi + \frac{2}{5}\pi = \frac{30}{15}\pi - \frac{20}{15}\pi +$$

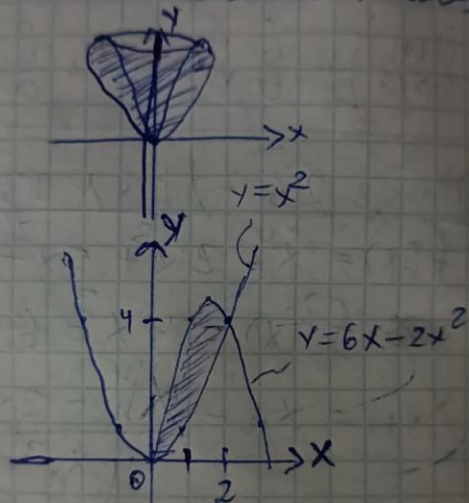
$$+ \frac{6}{15}\pi = 1\frac{1}{15}\pi \approx 1,07\pi$$

P364 / Ex 7

$$y = x^2$$

$$y = 6x - 2x^2$$

rotate about y-axis



$$x^2 = 6x - 2x^2$$

$$3x^2 = 6x$$

$$x(x-2) = 0$$

$$x_1 = 0, x_2 = 2$$

$$R = x, \text{ height} = 6x - 3x^2$$

$$\begin{aligned} & \int_0^2 (6x - 2x^2 - x^2) dx = \\ & \int_0^2 2\pi x (6x - 3x^2) dx = \\ & = \int_0^2 2\pi (6x^2 - 3x^3) dx = \\ & = 2\pi \left(\frac{6}{3} x^3 - \frac{3}{4} x^4 \right) \Big|_0^2 = \\ & = 2\pi \left(2 \cdot 2^3 - \frac{3}{4} \cdot 2^4 \right) = \\ & = \pi (2^5 - 3 \cdot 2^3) = \pi (32 - 24) = \\ & = 8\pi \end{aligned}$$

P 380 / E 25

$$f(x) = 1 + \sqrt{2+3x}$$

$$f(x) = y$$

$$y = 1 + \sqrt{2+3x}$$

$$y - 1 = \sqrt{2+3x}$$

$$y^2 - 2y + 1 = 2 + 3x$$

$$y^2 - 2y - 1 = 3x$$

$$x = \frac{y^2}{3} - \frac{2}{3}y - \frac{1}{3}$$

$$f^{-1}(x) = y = \frac{x^2}{3} - \frac{2}{3}x - \frac{1}{3}$$

P 381 / E 38

$$f(x) = 2x^3 + 3x^2 + 7x + 4$$

$$a = 4$$

$$f^{-1}(a) = ?$$

$$f^{-1}(a) = \frac{1}{f'(f^{-1}(a))}$$

$$2x^3 + 3x^2 + 7x + 4 = 4 = f(x)$$

$$x(2x^2 + 3x + 7) = 0$$

$$x_1 = 0$$

$$D = 9 - 4 \cdot 2 \cdot 7 < 0$$

$$f(0) = 4 \Rightarrow f^{-1}(4) = 0$$

$$f'(x) = 6x^2 + 6x + 7$$

$$f'(0) = 7$$

$$f^{-1}(4) = \frac{1}{f'(f^{-1}(4))}$$

$$f^{-1}(4) = \frac{1}{f'(0)}$$

$$f^{-1}(4) = \frac{1}{7}$$